

A REMARK ON EXPONENTIAL STABILITY OF TIME-DELAYED BURGERS EQUATION

YANBIN TANG AND MING WANG

School of Mathematics and Statistics
Huazhong University of Science and Technology
Wuhan, Hubei 430074, China

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ABSTRACT. In this paper, we consider the initial-boundary value problem of Burgers equation with a time delay. Using a fixed point theorem and a comparison principle, we show that the time-delayed Burgers equation is exponentially stable under small delays. The result is more explicit, but also complements, the result given by Weijiu Liu [Discrete and Continuous Dynamical Systems-Series B, 2:1(2002),47-56], which was based on the Liapunov function approach.

1. Introduction. The Burgers equation without a time delay has been the subject of numerous studies [2, 4, 6]. A simple derivation of Burgers equation occurs in the context of the kinematic-wave theory of traffic flow when the effect of vehicular diffusion is taken into account. According to the Fick's law of diffusion, the traffic flux is

$$q(x, t) = \rho(x, t)v(\rho(x, t)) - \nu\rho_x(x, t), \quad (1)$$

where $v(\rho(x, t)) = v_m(1 - \frac{\rho}{\rho_s})$ is the transport velocity, the axis is taken as one-dimensional highway, the traffic density $\rho = \rho(x, t)$ has the units of (vehicles)/(unit distance); the positive constant ρ_s is the saturation value of the density, where $\rho \in [0, \rho_s]$; the positive constants ν and v_m denote the diffusion coefficient and the maximum speed as $\rho \rightarrow 0$, respectively.

Substituting (1) into the continuity equation $\rho_t + q_x = 0$, we have the Burgers equation

$$\rho_t = \nu\rho_{xx} - v_m(1 - \frac{2\rho}{\rho_s})\rho_x. \quad (2)$$

Set $u(x, t) = v_m[1 - \frac{2\rho(x, t)}{\rho_s}]$, the Burgers equation (2) becomes

$$u_t(x, t) = \nu u_{xx}(x, t) - \frac{1}{2}[u^2(x, t)]_x.$$

The Burgers equation is a one-dimensional version of the Navier-Stokes equation, it would be the first step to understand the turbulence exhibited in a flow. Consider

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an ideal one-dimensional flow of fluid with the flow velocity $u = u(x, t)$. The rate of change of u following the fluid:

$$\frac{du}{dt} = \frac{d}{dt}u[x(t), t] = \frac{\partial u}{\partial t} + \frac{dx(t)}{dt} \frac{\partial u}{\partial x} = u_t(x, t) + u(x, t)u_x(x, t),$$

we also get the Burgers equation $u_t - \nu u_{xx} + uu_x = 0$.

What's the motivation of introducing a time delay into the Burgers equation? Just the same, we might have a delay τ to follow the fluid

$$\begin{aligned} \frac{du}{dt} &= \frac{d}{dt}u[x(t - \tau), t] = \frac{\partial u}{\partial t} + \frac{dx(t - \tau)}{dt} \frac{\partial u}{\partial x}(x(t - \tau), t) \\ &= u_t(x, t) + u(x, t - \tau)u_x(x, t), \end{aligned}$$

where $u(x(t - \tau), t)$ gives the velocity of the flow at position $x(t - \tau)$ at time t . This clearly shows how we obtain the time-delayed term in Burgers equation [7]:

$$u_t - \nu u_{xx} + u(x, t - \tau)u_x(x, t) = 0.$$

Consider the initial-boundary value problem of Burgers equation with a time delay

$$u_t - \nu u_{xx} + u(x, t - \tau)u_x(x, t) = 0, \quad (3)$$

$$u(0, t) = u(1, t) = 0, t > 0, \quad (4)$$

$$u(x, s) = \varphi(x, s), x \in [0, 1], s \in [-\tau, 0], \quad (5)$$

where $\nu > 0$ is the viscosity and $\tau > 0$ is the delay parameter, $\varphi(x, s)$ is an initial state in an appropriate function space. We know that some equations can be destabilized by small delays in the feedback no matter how small the delay is (see, e.g., [1, 3, 8, 9, 10, 14]). Does a small delay always destabilize Burgers equation? The answer is NO. In fact, by the Liapunov function and uniform Gronwall inequality, Weijiu Liu [7] shown that the solution of (3-5) is exponentially stable if the delay parameter τ is sufficiently small and gave an explicit estimate of the delay parameter in terms of the viscosity and initial condition. As we known, for a time-delayed equation, to construct a suitable Liapunov function is very difficult. Moreover, the estimates of time-delayed terms in Liapunov function are very technical.

In this paper, using a fixed point theorem and a comparison principle (see, e.g., [5, 11, 12, 13]), we get the exponential stability of time-delayed Burgers equation if delay is sufficiently small. The result of decay estimate of solution in space H_0^1 is more explicit, but also complements, the result given by Weijiu Liu[7], which was based on the Liapunov function approach.

2. Decay estimate of solution by Liapunov function. Define the linear operator A by

$$Aw = \nu w_{xx}, \quad D(A) = H^2(0, 1) \bigcap H_0^1(0, 1).$$

It's well known that A generates an analytic semigroup $T(t)$ on $L^2(0, 1)$ which is compact for $t > 0$, and

$$\|T(t)\| \leq ke^{-\omega t}, \quad \omega > 0, k \geq 1.$$

We also define the nonlinear operator $F : C([-\tau, 0], H_0^1(0, 1)) \rightarrow L^2(0, 1)$ by

$$F(\psi) = -\psi(-\tau)\psi_x(0), \quad \forall \psi \in C([-\tau, 0], H_0^1(0, 1)),$$

where F is locally Lipschitz.

In fact, $\forall \varphi, \psi \in C([- \tau, 0], H_0^1(0, 1))$, we have

$$\begin{aligned} & \|F(\varphi) - F(\psi)\|_{L^2} = \|\psi(-\tau)\psi_x(0) - \varphi(-\tau)\varphi_x(0)\|_{L^2} \\ & \leq \|\psi(-\tau)[\psi_x(0) - \varphi_x(0)]\|_{L^2} + \|\varphi_x(0)[\psi(-\tau) - \varphi(-\tau)]\|_{L^2} \\ & \leq \|\psi(-\tau)\|_{L^\infty} \|\psi_x(0) - \varphi_x(0)\|_{L^2} + \|\psi(-\tau) - \varphi(-\tau)\|_{L^\infty} \|\varphi_x(0)\|_{L^2}, \end{aligned}$$

$\forall u \in H_0^1(0, 1)$, by Sobolev Theorem and Poincaré inequality

$$\|u\|_{L^\infty}^2 \leq 2\|u\|_{L^2}\|u_x\|_{L^2} \leq \|u\|_{L^2}^2 + \|u_x\|_{L^2}^2 = \|u\|_{H_0^1}^2, \quad \|u_x\|_{L^2} \geq \pi\|u\|_{L^2},$$

we have

$$\begin{aligned} & \|F(\varphi) - F(\psi)\|_{L^2} \\ & \leq \|\psi(-\tau)\|_{H_0^1} \|\psi_x(0) - \varphi_x(0)\|_{L^2} + \|\psi(-\tau) - \varphi(-\tau)\|_{H_0^1} \|\varphi_x(0)\|_{L^2} \\ & \leq \|\psi\|_C \|\psi(0) - \varphi(0)\|_{H_0^1} + \|\varphi\|_C \|\psi(-\tau) - \varphi(-\tau)\|_{H_0^1} \\ & \leq (\|\psi\|_C + \|\varphi\|_C) \|\psi - \varphi\|_C, \end{aligned}$$

where

$$\|\varphi\|_C = \sup_{-\tau \leq s \leq 0} \|\varphi(\cdot, s)\|_{H_0^1}, \quad \forall \varphi(x, t) \in C([- \tau, 0], H_0^1(0, 1)).$$

Similarly, $\forall \varphi(x, t) \in C([- \tau, 0], H_0^1(0, 1))$, we have

$$\begin{aligned} \|F(\varphi)\|_{L^2} &= \|\varphi(-\tau)\varphi_x(0)\|_{L^2} \leq \|\varphi(-\tau)\|_{L^\infty} \|\varphi_x(0)\|_{L^2} \\ &\leq \|\varphi(-\tau)\|_{H_0^1} \|\varphi(0)\|_{H_0^1} \leq \|\varphi\|_C^2. \end{aligned}$$

Denote $u_t(s) = u(t + s)$, $s \in [- \tau, 0]$, then the Burgers equation (3) can be written in terms of the above operators as

$$\frac{du}{dt} = Au + F(u_t).$$

Therefore, IBVP(3-5) is transformed into the following integral equation

$$u(t) = \varphi(t), \quad t \in [- \tau, 0], \quad (6)$$

$$u(t) = T(t)\varphi(0) + \int_0^t T(t-s)F(u_s)ds, \quad t > 0. \quad (7)$$

The continuous solutions of (6-7) are called mild solutions of IBVP(3-5). Weijiu Liu [7] proved the existence and exponential decay of the global solution.

Theorem 2.1 ([7], P50). *For any initial condition $\varphi(x, t) \in C([- \tau, 0], H_0^1(0, 1))$, IBVP(3-5) has a unique global mild solution $u \in C([- \tau, \infty), H_0^1(0, 1))$.*

Theorem 2.2 ([7], P50). *For any initial condition $\varphi(x, t) \in C([- \tau, 0], H_0^1(0, 1))$, $\exists \tau_0 = \tau_0(\nu, \varphi)$, for $\tau < \tau_0$, the solution of IBVP(3-5) satisfies*

$$\|u_x(t)\| \leq \frac{k}{2} e^{-\omega t/2}, \quad \forall t \geq 0,$$

where k is a constant depending only on ν, τ and φ .

3. Decay estimate of solution by fixed point theorem.

Theorem 3.1. *For any nontrivial initial condition $\varphi(x, t) \in C([- \tau, 0], H_0^1(0, 1))$ such that $\|\varphi\|_C$ is sufficiently small, then $\exists \tau_0 = \tau_0(\nu, \varphi)$, for $\tau < \tau_0$, IBVP(3-5) has a unique global mild solution u on $[- \tau, \infty)$ with $u \in C([- \tau, \infty), H_0^1(0, 1))$. Furthermore, the following inequality*

$$\|u(\cdot, t)\|_{H_0^1(0,1)} \leq k\|\varphi\|_C(1 + \sqrt{t + \tau})e^{-\omega t}, \quad \forall t \geq -\tau,$$

holds true, therefore the trivial solution of IBVP(3-5) is asymptotically stable in $H_0^1(0, 1)$.

Proof of Theorem 3.1. 1) Define the set

$S = \{u \in C([- \tau, \infty), H_0^1(0, 1)) \mid u_0 = \varphi, \|u_t\|_C \leq k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega(t-\tau)}, \forall t \geq 0\}$, and the mapping G on S :

$$\begin{aligned} (Gu)(t) &= \varphi(t), \quad t \in [-\tau, 0], \\ (Gu)(t) &= T(t)\varphi(0) + \int_0^t T(t-s)F(u_s)ds, \quad t > 0. \end{aligned}$$

For $t \geq 0$ and $u \in S$, we have

$$\begin{aligned} &\|(Gu)(t)\|_{H_0^1(0,1)} \\ &\leq \|T(t)\varphi(0)\|_{H_0^1(0,1)} + \left\| \int_0^t (-A)^{\frac{1}{2}} T(t-s)F(u_s)ds \right\|_{L^2(0,1)} \\ &\leq ke^{-\omega t}\|\varphi(0)\|_{H_0^1(0,1)} + \int_0^t k(t-s)^{-\frac{1}{2}}e^{-\omega(t-s)}\|F(u_s)\|_{L^2(0,1)}ds \\ &\leq ke^{-\omega t}\|\varphi\|_C + \int_0^t k(t-s)^{-\frac{1}{2}}e^{-\omega(t-s)}\|u_s\|_C^2ds \\ &\leq ke^{-\omega t}\|\varphi\|_C + \int_0^t k^3\|\varphi\|_C^3(1 + \sqrt{s})^2(t-s)^{-\frac{1}{2}}e^{-\omega(t-s)}e^{-2\omega(s-\tau)}ds \\ &\leq ke^{-\omega t}\|\varphi\|_C + 2k^3\|\varphi\|_C^3e^{-\omega(t-2\tau)} \int_0^t (t-s)^{-\frac{1}{2}}(1+s)e^{-\omega s}ds. \end{aligned}$$

Denote

$$M = \sup_{[0, +\infty)} (1+s)e^{-\omega s},$$

then we have

$$\|(Gu)(t)\|_{H_0^1(0,1)} \leq ke^{-\omega t}\|\varphi\|_C[1 + 2k^2M\|\varphi\|_C^2e^{2\omega\tau} \int_0^t (t-s)^{-\frac{1}{2}}ds].$$

Choosing the sufficiently small $\|\varphi\|_C$ such that

$$4k^2M\|\varphi\|_C^2 < 1, \tag{8}$$

then there is a positive constant

$$\tau_1 = -\frac{1}{2\omega} \ln(4k^2M\|\varphi\|_C^2),$$

if $\tau \leq \tau_1$, we have

$$\|(Gu)(t)\|_{H_0^1(0,1)} \leq k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega t}, \quad t \geq 0. \tag{9}$$

For $t \in [-\tau, 0]$, it follows that $(Gu)(t) = \varphi(t)$, that is, $(Gu)_0 = \varphi$, and

$$\begin{aligned} \|(Gu)(t)\|_{H_0^1(0,1)} &= \|\varphi(t)\|_{H_0^1(0,1)} \leq \|\varphi\|_C \\ &\leq k\|\varphi\|_C(1 + \sqrt{t+\tau})e^{-\omega t}, t \in [-\tau, 0]. \end{aligned} \quad (10)$$

For $t \geq \tau$, (9) implies that

$$\begin{aligned} \|(Gu)_t\|_C &= \sup_{\theta \in [-\tau, 0]} \|(Gu)(t+\theta)\|_{H_0^1(0,1)} \\ &\leq \sup_{\theta \in [-\tau, 0]} k\|\varphi\|_C(1 + \sqrt{t+\theta})e^{-\omega(t+\theta)} \\ &\leq k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega(t-\tau)}, \quad t \geq \tau; \end{aligned}$$

for $0 \leq t \leq \tau$, (9) and (10) imply that

$$\begin{aligned} \|(Gu)_t\|_C &= \sup_{\theta \in [-\tau, 0]} \|(Gu)(t+\theta)\|_{H_0^1(0,1)} \\ &\leq \max\left\{ \sup_{-t \leq \theta \leq 0} \|(Gu)(t+\theta)\|_{H_0^1}, \sup_{-\tau \leq \theta \leq -t} \|(Gu)(t+\theta)\|_{H_0^1} \right\} \\ &\leq \max\{k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega(t-\tau)}, \|\varphi\|_C\} \\ &\leq k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega(t-\tau)}, \quad 0 \leq t \leq \tau, \end{aligned}$$

therefore, we have

$$\|(Gu)_t\|_C \leq k\|\varphi\|_C(1 + \sqrt{t})e^{-\omega(t-\tau)}, \quad t \geq 0.$$

Thus $G : S \rightarrow S$.

Due to (9) and (10), we also get

$$\|(Gu)(t)\|_{H_0^1(0,1)} \leq k\|\varphi\|_C(1 + \sqrt{t+\tau})e^{-\omega t}, \quad t \in [-\tau, \infty). \quad (11)$$

2) The set $\{(Gu)(t), u \in S\}$ is equicontinuous and GS is precompact.

The claim can be proved by the standard method. For example, the details are similar to that of Mengxing He [[5], P848-849], so they are omitted here.

3) The mapping $G : S \rightarrow S$ is continuous.

Denote

$$M_1 = \sup_{[0, +\infty)} (1 + \sqrt{s})e^{-\omega s},$$

choosing the sufficiently small $\|\varphi\|_C$ such that

$$2k^2 M_1 \|\varphi\|_C < 1, \quad (12)$$

then there is a positive constant

$$\tau_2 = -\frac{1}{\omega} \ln(2k^2 M_1 \|\varphi\|_C),$$

if $\tau \leq \tau_2$, we have $2k^2 M_1 \|\varphi\|_C e^{\omega\tau} < 1$.

For any fixed $T > 0, \forall \varepsilon > 0, \exists \delta = \sqrt{\frac{\omega}{\pi}} \varepsilon, \forall u_1, u_2 \in S$ such that

$$u_{10} = u_{20} = \varphi, \quad \sup_{0 \leq s \leq T} \|u_1 - u_2\|_{H_0^1} < \delta,$$

then $(Gu_1)(t) = (Gu_2)(t) = \varphi(t)$, $t \in [-\tau, 0]$ and $\forall t \in [0, T]$ we have

$$\begin{aligned}
& \| (Gu_1)(t) - (Gu_2)(t) \|_{H_0^1} \\
& \leq \| \int_0^t (-A)^{\frac{1}{2}} T(t-s) [F(u_{1s}) - F(u_{2s})] ds \|_{L^2(0,1)} \\
& \leq \int_0^t k(t-s)^{-\frac{1}{2}} e^{-\omega(t-s)} \|F(u_{1s}) - F(u_{2s})\|_{L^2} ds \\
& \leq \int_0^t k(t-s)^{-\frac{1}{2}} e^{-\omega(t-s)} (\|u_{1s}\|_C + \|u_{2s}\|_C) \|u_{1s} - u_{2s}\|_C ds \\
& \leq \int_0^t k(t-s)^{-\frac{1}{2}} e^{-\omega(t-s)} 2k \|\varphi\|_C (1 + \sqrt{s}) e^{-\omega(s-\tau)} \delta ds \\
& = 2k^2 M_1 \|\varphi\|_C e^{\omega\tau} \delta \int_0^t (t-s)^{-\frac{1}{2}} e^{-\omega(t-s)} ds \\
& \leq \delta \int_0^t p^{-\frac{1}{2}} e^{-\omega p} dp \leq \delta \int_0^{+\infty} p^{-\frac{1}{2}} e^{-\omega p} dp = \sqrt{\frac{\pi}{\omega}} \delta = \varepsilon.
\end{aligned}$$

Therefore, $G : S \rightarrow S$ is continuous.

According to the Schauder fixed point theorem, G has a fixed point in S , that is, $\exists u \in S$ such that $Gu = u$. This means that the problem (3-5) has a global mild solution $u \in C([-\tau, \infty); H_0^1(0, 1))$ which satisfies the inequality

$$\|u(\cdot, t)\|_{H_0^1} \leq k \|\varphi\|_C (1 + \sqrt{t + \tau}) e^{-\omega t}, \quad \forall t \in [-\tau, \infty).$$

The last inequality follows from (11).

4) Similar to the above step, there exists a small enough constant $\tau_3 > 0$, for the time delay $\tau \leq \tau_3$, we can prove the uniqueness of the solution.

Summarizing the above steps, denote $\tau_0 = \min\{\tau_1, \tau_2, \tau_3\}$, if $\tau \leq \tau_0$ and $\|\varphi\|_C$ is sufficiently small such that (8) and (12) hold true, then we prove Theorem 3.1. $\square$

4. Concluding remarks. Using the Liapunov function approach and the uniform Gronwall inequality, one can get the exponential decay of solution to the initial boundary value problem of the Burgers equation with a time delay, even if the estimate of time-delayed term in Liapunov function is very technical, see e.g. Weijiu Liu [7], etc. In this paper, using a fixed point theorem and a comparison principle, we show that the time-delayed Burgers equation is exponentially stable under small delays. The result is more explicit, but also complements, the result given by Weijiu Liu [7].

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REFERENCES

- [1] G. Avalos, I. Lasiecka and R. Rebarber, *Lack of time-delay robustness for stabilization of a structural acoustics model*, SIAM J. Control Optim., **37:5** (1999), 1394-1418.
- [2] A. Balogh and M. Krstić, *Burgers equation with nonlinear boundary feedback: H^1 stability, well-posedness and simulation*, Mathematical Problems in Engineering, **6** (2000), 189-200.
- [3] E. Beretta, F. Solimano and Y. B.Tang, *Analysis of a chemostat model for bacteria and virulent bacteriophage*, Discrete Continuous Dynam. Systems-B, **2** (2002), 495-520.

- [4] H. Choi, R. Temam, P. Moin and J. Kim, *Feedback control for unsteady flow and its application to the stochastic Burgers equation*, J. Fluid Mech., **253** (1993), 509–543.
- [5] M. X. He, *Global existence and stability of solutions for reaction diffusion functional differential equations*, J. Math. Anal. Appl., **199** (1996), 842–858.
- [6] W. J. Liu and M. Krstić, *Backstepping boundary control of Burgers equation with actuator dynamics*, Systems and Control Letters, **41** (2000), 291–303.
- [7] W. J. Liu, *Asymptotic behaviour of solutions of time-delayed Burgers equation*, Discrete Continuous Dynam. Systems-B, **2** (2002), 47–56.
- [8] H. Smith and P. Waltman, “The Theory of the Chemostat,” Cambridge University Press, Cambridge, UK, 1995.
- [9] Y. B. Tang and L. Zhou, *Stability switch and Hopf bifurcation for a diffusive prey-predator system with delay*, J. Math. Anal. Appl., **334** (2007), 1290–1307.
- [10] Y.B.Tang and L.Zhou, *Hopf bifurcation and stability of a competition diffusion system with distributed delay*, Publication of RIMS, **41** (2005), 579–597.
- [11] C. Travis and G. Webb, *Existence and stability for partial functional differential equations*, Trans. Amer. Math Soc., **204** (1974), 395–418.
- [12] C. Travis and G. Webb, *Partial differential equations with deviating arguments in the time variable*, J. Math. Anal. Appl., **56** (1976), 397–409.
- [13] C. Travis and G. Webb, *Existence, stability and compactness in the α - norm for partial functional differential equations*, Trans. Amer. Math Soc., **240** (1978), 129–143.
- [14] L. Zhou, Y. B. Tang and S. Hussein, *Stability and Hopf bifurcation for a delay competition diffusion system*, Chaos, Solitons and Fractals, **14** (2002), 1201–1225.

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E-mail address: tangyb@mail.hust.edu.cn

E-mail address: wangmingmaster@sina.com